

How Operators of Observables Emerge from Continuous Symmetries

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Hermitian operators and their physical observables are a fundamental principle of quantum mechanics, and yet their origins are commonly stated as a simple “promotion” of their non-operator counterparts. In this paper, we demonstrate an alternative method to arrive at these operators naturally from the continuous symmetries of our universe. Assuming basic knowledge in group theory, analysis, and topology, we build the mathematical tools relevant to this method. These mathematical concepts include: Lie groups and Lie algebras, the exponential map, basic functional analysis and spectral theory, and basic representation theory. While we exclude formal proofs, by structuring these topics in a natural and sequential manner, we can build a rather intuitive understanding of the mathematical structures. We then apply the method to a variety of symmetries to derive their associated Hermitian operators. Our derivations provide a geometric justification for the definition and properties of these operators. This approach is much more intuitive when compared to the postulate heavy canonical quantization.

I. INTRODUCTION

In most undergraduate quantum mechanics courses, the concepts are often presented in a postulate-heavy nature where we simply state the method of canonical quantization and define the mathematical constructs in a rather arbitrary manner. In fact, most courses are either introduced through wave-functions (see Griffiths [1]), or developed in a historical manner (see Liboff [2]).

This paper develops an alternative pedagogical technique to arrive at quantum operators as an alternative to canonical quantization. Due to its rather abstract nature, and heavy emphasis on mathematics, the presentation in this paper is unsuitable for a first-time student of quantum mechanics. However, by removing the unnecessary formal mathematical constructs, it is possible to apply the structure described in this paper to an undergraduate course. The traditional method at its core is still unintuitive while the method developed in this paper is much more natural.

This method is centered around the fact that symmetries in nature can correspond to operators in Hilbert spaces. We develop the mathematical tools in a linear manner, without any burden of proof to maintain conciseness. Throughout the paper, I will assume the reader has a basic understanding of linear algebra, group theory, topology, and analysis.

This paper will start with Lie groups and Lie algebras, where the important properties and theorems are naturally developed as we work towards the exponential map. We then develop the idea of spectra and state the Spectral theorem in its multiplicative operator form. We then define representations and describe how they connect Lie groups to operators through Stone’s theorem on one parameter unitary groups.

II. SYMMETRIES AND LIE GROUPS

In order to properly discuss symmetries, we must first understand the mathematical language in which they are represented: Lie groups. However, to build the intuition for Lie groups, we must discuss the idea of manifolds first.

A. Manifolds and Tangent Spaces

Generally stated, a manifold is an n -dimensional space that locally resembles $\mathbb{R}^n$. We can first define the most general manifold:

Definition 1. Given a topological space M , we call M a **Topological Manifold** of dimension n if it is:

- (a) A Hausdorff Space: for every pair of distinct points $p, q \in M$, there exist disjoint open subsets $U, V \subseteq M$ such that $p \in U$ and $q \in V$.
- (b) Second Countable: there exists a countable basis for the topology of M .
- (c) Locally Euclidean of dimension n : for every point in M there exists a neighborhood which is homeomorphic to an open subset of $\mathbb{R}^n$. [3]

Essentially, an n -dimensional manifold must be "well behaved" and be isomorphic to $\mathbb{R}^n$ for some small enough area around each point. Some examples of manifolds include the unit spheres S^n , a Klein bottle, curved spacetime, finite-dimensional real vector spaces, and the general linear group $\mathbf{GL}(n, \mathbb{C})$. However, we wish to impose an additional structure on our manifolds such that we can in a way "do calculus" on these manifolds.

Definition 2. A **Smooth Manifold** is a topological manifold M , together with a smooth structure¹ on M . [3]

Smooth manifolds bring the differentiable structure of $\mathbb{R}^n$ onto these arbitrary spaces in the sense that you can define functions with derivatives and integrals on the manifold. In fact, this smooth structure gives the space one more important structure: the tangent space.

Definition 3. Given a manifold M and a point $p \in M$, a **Derivation** is a linear map $v : C^\infty(M) \rightarrow \mathbb{R}$ such that it follows the Leibniz identity, that is, $v(fg) = v(f) \cdot g(p) + f(p) \cdot v(g)$ for all $f, g \in C^\infty(M)$. [3]

where $C^\infty(M)$ is the ring of smooth functions over M . This idea generalizes the derivative algebraically, to be a functional that satisfies the product rule. This algebraic generalization then allows us to construct the tangent space:

Definition 4. The set of all derivations of $C^\infty(M)$ at a point $p \in M$ is called the **Tangent Space** at p . We denote this set as $T_p M$. [3]

This can also be thought of as the derivative of all curves passing through the point $p \in M$. As the name suggests, this is the generalization of tangent lines and tangent planes for curves in $\mathbb{R}^2$ and surfaces in $\mathbb{R}^3$ respectively. We shall conclude this section with two useful theorems about smooth manifolds.

¹ a maximal smooth atlas

Proposition 1. *The tangent space $T_p M$ is a vector space with dimension $\dim(M)$.*

Proposition 2. *Let M be a smooth manifold and X a topological space. If there exists a homeomorphism $f : X \rightarrow M$, then X is a topological manifold and there exists a smooth structure on X . Furthermore, f is a C^∞ -diffeomorphism².*

The second proposition essentially allows us to transfer the smooth structure from one manifold to another, which we will use in the following sections to illustrate the smoothness of certain manifolds.

B. Lie Groups and Lie Algebras

Definition 5. A **Lie Group** is a smooth manifold G , that also is a group algebraically, where the group product and inverse are smooth maps. [4]

This definition is rather abstract, and hard to interpret without an example. A simple illustration of this can be seen by taking the unit circle:

$$S^1 = \{z \in \mathbb{C} : |z| = 1\}$$

It is clear that this is in fact a manifold, and one can convince oneself that it has a smooth structure³. However, one can also view this as a group algebraically by taking each point on the circle as an element in the group. This becomes clearer if we denote each element as $z = e^{ix}$ for $x \in [0, 2\pi)$. Then any product of these elements clearly remain in the group, and the rest of the group axioms are trivially true by taking $z = 1$ as the identity element. And thus, this construction is an example of a Lie group.

Definition 6. A **Matrix Lie Group** is a closed subgroup⁴ of the general linear group $G \subseteq \mathbf{GL}(n, \mathbb{C})$. [4]

In fact, most Lie groups are isomorphic to a matrix Lie group, and the ones that we will work with in this paper are all under this category. A quick example of a matrix Lie group is $U(n)$, the set of all $n \times n$ unitary matrices. Notice that in the one dimensional case, $U(1)$, elements have the form

$$U(1) = \{[e^{ix}] \in \mathbf{GL}(1, \mathbb{C}) : x \in [0, 2\pi)\}$$

In fact, this is exactly homeomorphic to the smooth manifold S^1 , by simply mapping the element of the matrix to itself. Then by proposition 2, $U(1)$ is indeed a smooth manifold. The group structure of $U(1)$ is clear and thus $U(1)$ is indeed a Lie group.

To develop the idea of a Lie algebra, we first give its abstract definition before motivating them through Lie groups.

Definition 7. A **Lie algebra** is a vector space $\mathfrak{g}$ over a field F together with a binary operation $[\cdot, \cdot] : \mathfrak{g} \times \mathfrak{g} \rightarrow \mathfrak{g}$ such that:

- (a) $[\cdot, \cdot]$ is bilinear
- (b) $[\cdot, \cdot]$ is skew symmetric, i.e. $[X, Y] = -[Y, X]$ for all $X, Y \in \mathfrak{g}$

² a bijective map where both f and f^{-1} are C^∞

³ The proof of this is rather burdensome through charts and atlases, and thus, we omit the proof.

⁴ G is a closed subset of $\mathbf{GL}(n, \mathbb{C})$

(c) The Jacobi Identity holds, i.e

$$[X, [Y, Z]] + [Y, [Z, X]] + [Z, [X, Y]] = 0 \quad (2.1)$$

for all $X, Y \in \mathfrak{g}$.

We call this operation the **Lie Bracket**. We say two elements $X, Y \in \mathfrak{g}$ **commute** if $[X, Y] = 0$. [4]

Notice that these properties are exactly those of the commutators in quantum mechanics. Indeed the quantum commutator is an example of a Lie bracket. As the names indicate, there exists a connection between the idea of a Lie group and a Lie algebra.

Proposition 3. *Every Lie group G has an associated Lie algebra $\text{Lie}(G)$.*

There is a method as outlined in [3] to construct a Lie algebra (and a Lie bracket) out of any general Lie group, however, the method is out of the scope of this paper. If we narrow our focus to matrix Lie groups, we can build a simpler method. But first, we must discuss the precise association between the Lie group and the Lie algebra based upon the general method.

Theorem 4. *Let G be a Lie group. Then there exists an isomorphism between the associated Lie algebra $\text{Lie}(G)$ and the tangent space at the identity $T_e M$. [3]*

Then, by finding every tangent vector at the identity, we find all the elements of the associated Lie algebra. In the context of matrix Lie groups, we find all the tangent vectors at the identity I_n .

To construct this association, let us start by defining a Lie algebra denoted $\mathfrak{gl}(n, \mathbb{C})$, by endowing the set of all $n \times n$ matrices $M_n(\mathbb{C})$ with the commutator bracket $[A, B] = AB - BA$ for $A, B \in M_n(\mathbb{C})$. Since $\mathbf{GL}(n, \mathbb{C})$ is an open subset⁵ of $\mathfrak{gl}(n, \mathbb{C})$, there must be an isomorphism between $T_I \mathbf{GL}(n, \mathbb{C})$ and $\mathfrak{gl}(n, \mathbb{C})$. [3] One can be convinced of this by imagining velocity curves through the identity. Since the set is open, then there exists a neighborhood around I where the curves can pass through. Then, by proposition 1 tangent space is a vector space with the same dimension as $\mathbf{GL}(n, \mathbb{C})$ and with the same structure of $\mathfrak{gl}(n, \mathbb{C})$, which is exactly $\mathfrak{gl}(n, \mathbb{C})$.

Since all matrix Lie groups are subsets of $\mathbf{GL}(N, \mathbb{C})$, we expect the Lie algebra of a matrix Lie group to be a subset of $\mathfrak{gl}(N, \mathbb{C})$ with the commutator as the Lie bracket. We state this relationship as a theorem:

Theorem 5. *Let G be a (closed) matrix Lie group. The associated Lie algebra of G , denoted $\mathfrak{g}$ is defined as:*

$$\mathfrak{g} = \{X \in \mathfrak{gl}(n, \mathbb{C}) : e^{tX} \in G, \forall t \in \mathbb{R}\} \quad (2.2)$$

One can imagine the exponential map wrapping the vector from the flat tangent space on to the curved manifold. In fact if one imagines taking the derivative of the exponential at the origin, we get exactly the Lie algebra element, meaning that near the identity, the Lie group and the Lie algebra behave the same. This is exactly how the tangent space behaves.

Then let us collect some important properties of the Lie algebra that we discovered during this construction.

Theorem 6. *Given a Lie group G :*

(a) *There exists an associated Lie algebra $\mathfrak{g}$.*

⁵ Since the determinant is a polynomial function, it is continuous. Also since $GL(n, \mathbb{C})$ consists of matrices with determinant not equal to 0, then the range of the determinant is $\mathbb{C} \setminus \{0\}$, an open set. We know that if f is continuous then the pre-image of an open set is open, then $\mathbf{GL}(n, \mathbb{C})$ must be an open set

- (b) $\mathfrak{g}$ is the tangent space at the identity of G .
- (c) The dimension of $\mathfrak{g}$, $T_p G$, and G are equal for any $p \in G$.
- (d) $\mathfrak{g} \cong T_e G$.
- (e) If G is a matrix Lie group:
 - (i) $\mathfrak{g}$ is a sub-algebra of $\mathfrak{gl}(n, \mathbb{C})$.
 - (ii) The Lie bracket is the commutator.
 - (iii) $\mathfrak{g} = \{X \in \mathfrak{gl}(n, \mathbb{C}) : e^{tX} \in G, \forall t \in \mathbb{R}\}$

Again, looking at $U(1)$ as an example, the tangent space at the identity is simply the vertical line consisting of all imaginary numbers. This corresponds exactly with the algebraic definition:

$$\mathfrak{u}(1) = \{X \in \mathfrak{gl}(n, \mathbb{C}) : e^{tX} \in U(1), \forall t \in \mathbb{R}\} = \{ix : x \in \mathbb{R}\}$$

III. FUNCTIONAL ANALYSIS AND OPERATORS

Let us ensure that we have a strong understanding of operators in quantum mechanics. In this section we will review some definitions and precisely state the properties of these operators.[5]

Essentially, functional analysis generalizes the ideas of linear algebra to infinite dimensional spaces. The most important idea is the concept of eigenvalues and eigenfunctions. We want to work in infinite dimensions, because for operators such as the kinematic operators, which have an infinite set of "eigenvalues", the dimension of the space must be infinite.

A. Hilbert Spaces and Operators

Let us state some definitions that are integral to our discussion of operators:

Definition 8. A **Normed Space** is a vector space on which a norm is defined.

Definition 9. A **Banach space** is a complete normed space.

Definition 10. A **Hilbert space** is a Banach space where the norm is defined through an inner product.[6]

Definition 11. If V, W are normed spaces, an operator $T : V \rightarrow W$ is called **Bounded** if $\sup_{\psi \in V \setminus \{0\}} \frac{\|T\psi\|}{\|\psi\|} < \infty$. Let us call this supremum the **Operator Norm**. [6]

Definition 12. If V is a normed space, the collection of all bounded linear functionals of V is called the **Dual Space** to V , denoted V^* . [6]

A quick note on Hilbert spaces and quantum mechanics, we use the bra-ket notation, because, for every vector in the dual space of V , there exists a vector in the original space $u \in V$ where the functional is $f_u(v)$ acting on any $v \in V$ as the inner product. This is guaranteed by the Riesz representation theorem.

Now let us continue to discuss some important properties of operators.

Definition 13. Let $\mathcal{B}(\mathcal{H})$ denote the set of all bounded operators $A : \mathcal{H} \rightarrow \mathcal{H}$

Theorem 7. Given a bounded operator $A \in \mathcal{B}(\mathcal{H})$, there exists a unique linear operator $A^\dagger : \mathcal{H} \rightarrow \mathcal{H}$ such that $\langle \phi | A \psi \rangle = \langle A^\dagger \phi | \psi \rangle$ called the **Hermitian Adjoint**. For unbounded operators, the adjoint acts on $\text{Dom}(A^\dagger)$ which is defined as the elements ψ such that $\langle A \phi | \psi \rangle = \langle \phi | \eta \rangle$ for all $\phi \in \text{Dom}(A)$ ⁶ where we define $A^\dagger \psi = \eta$. [6]

⁶ equivalent to stating that $\langle A \phi | \psi \rangle$ is bounded for all $\phi \in \text{Dom}(A)$

Definition 14. An operator⁷, A is called **Symmetric** if $\langle A\phi|\psi\rangle = \langle\phi|A\psi\rangle$ and called **Self-Adjoint** if $\text{Dom}(A) = \text{Dom}(A^\dagger)$ and $A = A^\dagger$.⁸[6]

In quantum mechanics, we typically refer to both symmetric and self-adjoint operators as Hermitian. In a strict sense however, Hermitian should refer to self-adjoint operators. If the operator is bounded, $\text{Dom}(A) = \mathcal{H}$, and thus symmetric operators are self-adjoint.

Definition 15. A symmetric operator A is called **Essentially Self-Adjoint** if $A^\dagger + iI$ and $A^\dagger - iI$ are injective on $\text{Dom}(A^\dagger)$.

Proposition 8. *The closure of an essentially self-adjoint operator is self-adjoint*

In general, an essentially self-adjoint operator is as good as a self-adjoint operator since we can just extend its domain by taking its closure.

Definition 16. A bounded operator $U \in \mathcal{B}(\mathcal{H})$ is **Unitary** if $UU^\dagger = U^\dagger U = I$, where I is the identity operator on $\mathcal{H}$ [6]

Unitary operators are important in quantum mechanics because they preserve the inner product. Since probabilities and expectation values are based upon the inner product, unitary operators make sure probabilities are conserved.

B. Spectral Theory

In this subsection, we generalize the idea of eigenvectors and eigenvalues to infinite dimensional spaces. We will focus on motivating our results through bounded operators since they are more well behaved, but we will see that our results generalize pretty smoothly into unbounded operators.

Definition 17. Given a bounded operator $A \in \mathcal{B}(\mathcal{H})$, the **Resolvent Set** of A is the set of all $\lambda \in \mathbb{C}$ such that the operator $(A - \lambda I)$ has a bounded inverse. **The Spectrum** of A , denoted by $\sigma(A)$, is the complement of the resolvent set in $\mathbb{C}$. [6]

When we find eigenvalues of an operator A in finite dimensional spaces, we find all λ such that $\det(A - \lambda I) = 0$. Notice that this is equivalent to the definition of the spectrum restricted to finite dimensional spaces. Then we can say that the spectrum is a generalized set of eigenvalues.

Theorem 9 (Spectral Theorem for Bounded Operators). *Suppose $A \in \mathcal{B}(\mathcal{H})$ is self-adjoint. Then there exists a measure space (X, Σ, μ) , a real-valued function λ on X , and a unitary map $U : \mathcal{H} \rightarrow L^2(X, \mu)$ such that*

$$[UAU^{-1}(\psi)](x) = \lambda(x)\psi(x) \quad (3.1)$$

for all $\phi \in L^2(X, \mu)$ [6]

This result shows us that the spectrum of an operator $A \in \mathcal{B}(\mathcal{H})$ behaves like a set of eigenvalues. However, as the kinematic observables are unbounded operators, it is important that we generalize these results to unbounded operators.

Definition 18. Let A be an unbounded operator on $\mathcal{H}$. $\lambda \in \mathbb{C}$ is in the **Resolvent set** of A if there exists a bounded operator B such that:

⁷ the domain must also be dense in $\mathcal{H}$

⁸ typically for bounded operators $\text{Dom}(A) = \mathcal{H}$, which makes these two conditions equivalent.

- (a) for all $\psi \in \mathcal{H}$, $B\psi \in \text{Dom}(A)$ and $(A - \lambda I)B\psi = \psi$
- (b) for all $\psi \in \text{Dom}(A)$, $B(A - \lambda I)\psi = \psi$

If no such bounded operator exists, then λ is in the **Spectrum** of A . [6]

This is a simple generalization, but with additional domain restrictions on the inverse. Generalizing the spectral theorem:

Theorem 10 (General Spectral Theorem). *Given a self-adjoint operator A on $\mathcal{H}$, there exists a measure space, (X, Σ, μ) , a real valued function λ on X and a unitary map $U : \mathcal{H} \rightarrow L^2(X, \mu)$ such that:*

$$U(\text{Dom}(A)) = \{\psi \in L^2(X, \mu) : h\psi \in L^2(X, \mu)\} \quad (3.2)$$

and

$$[UAU^{-1}(\psi)](x) = \lambda(x)\psi(x) \quad (3.3)$$

for all $\psi \in U(\text{Dom}(A))$ [6]

Proposition 11. *If $A : \mathcal{H} \rightarrow \mathcal{H}$ is self-adjoint, then the spectrum $\sigma(A) \subseteq \mathbb{R}$. [6]*

This will be important when we discuss these self-adjoint operators in the context of quantum mechanics. Let us now generalize the idea of eigenspaces to **Spectral Subspaces**. The formal definition is rather tedious, so let us restrict our view to their properties.

Theorem 12. *For each Borel set⁹ $E \subseteq \sigma(A)$, there exists a subspace V_E of $\mathcal{H}$ such that:*

- (a) $V_{\sigma(A)} = \mathcal{H}$
- (b) $V_{\emptyset} = \{0\}$
- (c) *If E and F are disjoint, then $V_E \perp V_F$*
- (d) *For any E , $V_E \cap \text{Dom}(A)$ is invariant under A*
- (e) *If $E \subset [\lambda - \epsilon, \lambda + \epsilon]$ and $\psi \in V_E$, then $\|(A - \lambda I)\psi\| \leq \epsilon\|\psi\|$ [6]*

Translating this idea into the language of eigenvectors, we imagine each V_E as the span of an eigenvector associated with E . Property (a) captures fact that a complete set of eigenvectors form a basis for the vectors space. Property (c) captures the fact that eigenvectors are orthogonal to each other. Property (d) represents the fact that A acting on a vector in V_E returns a vector in the same direction, while property (e) claims that the eigenvectors are not exact. It is helpful to imagine E as a "smeared" eigenvalue, with a "smeared" eigenvector ψ . This idea can be seen clearer through the following theorem:

Theorem 13. *If A is a self-adjoint operator on $\mathcal{H}$, then $\lambda \in \mathbb{R}$ belongs to the spectrum of A if and only if there exists a sequence $\{\psi_n\}_{n=1}^{\infty}$ such that: [6]*

$$\lim_{n \rightarrow \infty} \frac{\|(A - \lambda I)\psi_n\|}{\|\psi_n\|} = 0 \quad (3.4)$$

This theorem essentially states that ψ_n are approximate eigenvectors for λ an approximate eigenvalue.

⁹ One can think of Borel sets as being generated by the arbitrary union and intersection of open sets in $\mathbb{R}$

C. Stone's Theorem for One Parameter Unitary Groups

To end this section, we shall world towards the theorem at the heart of this paper.

Definition 19. A **One-Parameter Unitary Group** on G is a family of unitary operators $U(t) : \mathbb{R} \rightarrow \mathcal{H}$ such that $U(0) = I$ and that $U(s+t) = U(s)U(t)$ for all $s, t \in \mathbb{R}$. [6]

Note that despite its name, these families are not groups but a collection of operators. In general a one-parameter group is a collection of group homomorphisms, $\phi : \mathbb{R} \rightarrow G$, where we have taken a unitary operator to be our homomorphism and $\mathcal{H}$ as our group. It has the name of a group because the family preserves the group structure.

Definition 20. We say a one parameter unitary group is **strongly continuous** if: [6]

$$\lim_{s \rightarrow t} \|U(t)\psi - U(s)\phi\| = 0 \quad (3.5)$$

Theorem 14 (Stone's Theorem for One Parameter Unitary Groups). *Suppose $\{U(t) : t \in \mathbb{R}\}$ is a strongly continuous one-parameter unitary group on $\mathcal{H}$. Then the infinitesimal generator A of $\{U(t)\}$ is densely defined, self-adjoint, and $U(t) = e^{itA}$ for all $t \in \mathbb{R}$. [6]*

The most important takeaway from this, is that $\{U(t)\}_{t \in \mathbb{R}}$ is generated by the self-adjoint operator A through the given relation.

IV. REPRESENTATION THEORY

This section acts as the bridge between sections II and III. Similarly to the previous section, we will discuss finite dimensional representations first since they are easier to motivate, before moving onto infinite dimensional representations.

Definition 21. The general linear group of a vector space V is the group of all automorphisms on V . We denote this as $\mathbf{GL}(V)$. Similarly we define the Lie algebra $\mathfrak{gl}(V)$ as the set of all endomorphisms on V with the commutator as its Lie bracket.

Definition 22. Given a Lie group G , a **Finite Dimensional Representation** of G is a smooth group homomorphism $\Pi : G \rightarrow \mathbf{GL}(V)$, where V is a finite dimensional vector space. [6]

Definition 23. Given a Lie algebra $\mathfrak{g}$, a **Finite Dimensional Representation** of $\mathfrak{g}$ is a linear map $\pi : \mathfrak{g} \rightarrow \mathfrak{gl}(V)$, where V is a finite dimensional vector space. Given an associated Lie group G and a representation, we can find the Lie algebra representation by defining $\Pi(e^X) = e^{\pi(X)}$. [6]

These maps are what encode the group action of the Lie groups onto a vector space. Then, since the group action is represented by an element of $\mathbf{GL}(V)$, it is clear that the group action is represented by an operator on V .

Definition 24. Given a finite dimensional complex Hilbert space $\mathcal{H}$, denote $U(\mathcal{H})$ as the group of all unitary Operators on $\mathcal{H}$. A finite-dimensional **Unitary Representation** of a Lie group G is a smooth group homomorphism $\Pi : G \rightarrow U(\mathcal{H})$. [6]

For infinite dimensions, we shall focus on the case of unitary representations, since those are the operators of interest.

Definition 25. We say a representation is **Strongly Continuous** if:[6]

$$\lim_{n \rightarrow \infty} \|\Pi(A_n)\psi - \Pi(A)\phi\| = 0 \quad (4.1)$$

This is the same definition as the strong continuity of single parameter Unitary groups, if we restrict our operators to be parametrized by $\mathbb{R}$ and take the image of Π . Note that every finite-dimensional Unitary Representation is automatically continuous

Definition 26. Given a Hilbert space $\mathcal{H}$ Suppose G is a Lie group, then a **Unitary Representation** of G is a strongly continuous homomorphism $\Pi : G \rightarrow U(\mathcal{H})$. [6]

Proposition 15. *Every Lie group has a unitary representation on a Hilbert Space*

Now, we may construct the method through which we motivate our Hermitian observables. First let G be a Lie group, and $\mathfrak{g}$ be its associated Lie algebra. If we fix an $X \in \mathfrak{g}$, then e^{tX} is a one-parameter subgroup of G .

Then take the unitary representation of this one-parameter subgroup e^{tX} on a Hilbert space $\mathcal{H}$. As a function of t , this forms a strongly continuous one-parameter unitary group $\Pi(e^{tX})$. Then by Stone's theorem on one-parameter unitary groups, there exists a self-adjoint operator A such that

$$\Pi(e^{tX}) = e^{itA}$$

for all $t \in \mathbb{R}$. Then by taking the associated Lie algebra for the Lie group, we get the following:

Theorem 16. *Given a Lie group G with a unitary representation Π , the associated Lie algebra has a representation of the form $\pi(X) = iA$ for some self-adjoint operator A for any $X \in \mathfrak{g}$. Furthermore, the Lie bracket is conserved when taking the representation of a Lie algebra. [6]*

This theorem is a direct result of the above construction and the definition of the representation of the associated Lie algebra. This is the exact construction that we will use to motivate the link between Hermitian operators and observables in quantum mechanics.

V. SYMMETRIES IN QUANTUM MECHANICS

If we wish to apply theorem 16 in the context of quantum mechanics, we must alter the theorem slightly.

Theorem 17. *Given a Lie group G with a unitary representation Π , the associated Lie algebra has a representation of the form $\pi(X) = \frac{i}{\hbar}B$ for some self-adjoint operator B for any $X \in \mathfrak{g}$. Furthermore, the Lie bracket is conserved when taking the representation of a Lie algebra. [6]*

One can imagine $\hbar$ as a factor to match the dimensions of the operators. The scaling of the operator does not change the mathematics of this scheme, but (unfortunately), we must treat as a postulate to encode the non-commutativity of our operators. and so we may follow the scheme described below:

1. Start with a quantum system with a continuous symmetry, and determine its Hilbert space.
2. Find the Lie group that encodes this symmetry, and find the associated Lie algebra for the group and its generators
3. Use theorem 17 to find a unitary representation of the Lie group and the Lie algebra with its Lie bracket.

In the last step, the unitary operation is the physical symmetry that we encoded using the Lie group, and the self-adjoint operator is the physical observable that is the conserved quantity associated with the physical symmetry. This method provides a more intuitive way of constructing the self-adjoint operators associated with physical observables. We shall now apply this scheme to some examples of symmetries, starting with a finite dimensional case.

A. Rotational Symmetry: $SU(2)$

For this section, we shall simply postulate the existence of spin for the sake of conciseness. One may motivate this through the existence of half integer angular momentum, or by the fact that $SU(2)$ is the double cover of $SO(3)$, but we omit this for the sake of demonstrating scheme constructed in this paper.

We choose the group $SU(2)$ because it is indeed the group that encodes rotational symmetry in the most general manner. This group is the group of 2×2 unitary matrices with determinant 1. To see the manifold structure, one may imagine it as S^3 , the surface of a 4-dimensional sphere. Its Lie algebra is generated by the skew-Hermitian Pauli matrices and satisfy the Lie bracket:

$$\left[\frac{i\sigma_i}{2}, \frac{i\sigma_j}{2} \right] = -\epsilon_{ijk} \frac{\sigma_k}{2} \quad (5.1)$$

Let σ_z generate a one-parameter unitary group on the Hilbert space. Then $\Pi(e^{it\frac{\sigma_z}{2}}) = e^{it\frac{B}{\hbar}}$ is a unitary representation of this group, where by theorem 17, $\pi(i\frac{\sigma_z}{2}) = \frac{i}{\hbar}B$ is a representation of the Lie algebra, with B a self-adjoint operator.

Relabeling into more familiar notation we have $U(t, \hat{z}) = e^{-itS_z}$, where we introduce a negative in the exponent to follow the standard convention. Repeating the same steps for the x and y components, we get that $U(t, \hat{y}) = e^{-itS_y}$ and $U(t, \hat{x}) = e^{-itS_x}$ where S_x, S_y, S_z are self-adjoint operators.

Furthermore, we know that $\pi(\frac{\sigma_k}{2})$ must satisfy the commutation relations given by eq. (5.1), and thus $[\frac{i}{\hbar}S_i, \frac{i}{\hbar}S_j] = -\frac{1}{\hbar}\epsilon_{ijk}S_k$ which implies that:

$$[S_i, S_j] = i\hbar\epsilon_{ijk}S_k$$

Since this manifold is compact, we expect it to have a non-trivial representation on a finite dimensional Hilbert space. As an example, let us take the two-dimensional Hilbert space representing a spin $\frac{1}{2}$ system. Since all finite dimensional Hilbert spaces are isomorphic to $\mathbb{C}^n$, then a valid representation of our operators are just the Pauli matrices themselves. Then $\pi(i\frac{\sigma_k}{2}) = i\frac{\sigma_k}{2} = \frac{i}{\hbar}S_k$, giving us that

$$S_k = \frac{\hbar}{2}\sigma_k$$

This easily generalizes to n -dimensions by simply taking a representation of the Pauli matrices in the form of an $n \times n$ matrix. Then for spin, we have found the Hermitian operators that are associated with the physical observable of spin. This example was a good way to introduce the method, however, due to the abstract nature of spin, it does not serve as a good illustration of the connection between symmetries to operators.

From here, we may discuss the spectrum and the spectral subspaces of these operators. By proposition 11 we know that the spectrum of self-adjoint operators are real, and since S_k are self adjoint operators representing the spin angular momentum of the system, we expect its spectrum to

be the possible values of the spin. Since this operator is bounded and acting on a finite dimensional space, the spectrum is exactly the eigenvalues of the spin matrices.

For physical rotations in $\mathbb{R}^3$, we expect the same rotational symmetry to hold. We also know that the associated conserved quantity to rotations in $\mathbb{R}^3$ to be orbital angular momentum. Then if we represent $SU(2)$ to an infinite dimensional Hilbert space, we can get the Hermitian operators associated with orbital angular momentum. Since our above construction was general up to where we looked at the two-dimensional case, by relabeling our operators S_k into L_k we know the unitary representation of physical rotations have the form $U(t, \hat{z}) = e^{-itL_z}$, $U(t, \hat{y}) = e^{-itL_y}$, and $U(t, \hat{x}) = e^{-itL_x}$, where L_x , L_y and L_z are self-adjoint operators satisfying the commutation relations: $[L_i, L_j] = i\hbar\epsilon_{ijk}L_k$.

Since we are applying these rotations to the Hilbert space, we can apply the usual ladder method of achieving the differential operator representation of the angular momentum operators in $L^2(\mathbb{R}^3, \mathbb{C})$ through ladder operators. In this case, however, the orbital angular momentum operators are not bounded, and they act on an infinite dimensional Hilbert space. Then we must think of its spectrum as generalized eigenvalues and its spectral subspace as generalized eigenvectors. This introduces some pathologies of the theory that we will discuss in section V C.

B. Translational Symmetry: $\mathbb{R}^3$ and $\mathbb{R}$

The Lie group that represents translations in space is $\mathbb{R}^3$ with vector addition as the group operation. We can imagine the manifold of $\mathbb{R}^3$ to be exactly itself since $\mathbb{R}^3$ obviously resembles $\mathbb{R}^3$ locally. Further, the Lie algebra of $\mathbb{R}^3$ is exactly $\mathbb{R}^3$ again, and is generated by the basis vectors $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$. From this we know that the Lie algebra is abelian, and thus

$$[\mathbf{e}_i, \mathbf{e}_j] = 0$$

Since the Lie group is not compact, there does not exist a finite dimensional representation, and thus we must work with infinite dimensional representations. Particularly, we work with the infinite dimensional Hilbert space.

Let $\mathbf{e}_x$ generate a one-parameter unitary group, such that $\Pi(e^{t\mathbf{e}_x}) = e^{it\frac{B}{\hbar}}$. Further by theorem 17, $\pi(\mathbf{e}_x) = \frac{i}{\hbar}B$ for a self adjoint operator B . Renaming these operators into more familiar notation, we have $T(t, \hat{x}) = e^{-itp_x}$, where we have again introduced a negative sign in the exponent to follow the standard convention. Similarly $T(t, \hat{y}) = e^{-itp_y}$, and $T(t, \hat{z}) = e^{-itp_z}$. Then, the operators must satisfy the commutation relations $[\frac{i}{\hbar}p_i, \frac{i}{\hbar}p_j] = 0$, which implies that,

$$[p_i, p_j] = 0$$

To make our representation more concrete, let us analyze how the translational operator acts infinitesimally on wave functions in $L^2(\mathbb{R}^3, \mathbb{C})$, since all separable infinite dimensional Hilbert spaces are isomorphic to $L^2(\mathbb{R}^3, \mathbb{C})$. Let $T(a, \hat{x})$ act on the wave function $\psi(x)$ such that

$$T(a, \hat{x})\psi(x) = \psi(x - a) = \psi(x) - a\partial_x\psi(x) + \dots$$

Also taking the expansion of $e^{\frac{i}{\hbar}ap_x}$, we get

$$e^{\frac{i}{\hbar}ap_x} = I - \frac{i}{\hbar}ap_x + \dots$$

and by matching terms, we find that

$$p_x = -i\hbar\partial_x$$

We can conduct a similar scheme on $\mathbb{R}$ in one dimension to explore time-translation symmetry. Then, we find there exists a self-adjoint operator such that it generates time evolution, which we know is the Hamiltonian, since that is the conserved quantity associated with time-translational symmetry. Again since both of these operators are unbounded operators acting on infinite dimensional Hilbert spaces with continuous spectra, there exist certain pathologies when discussing the spectrum and eigenkets of the operators.

C. Further Generalizations

In the previous subsections, we have explored the general scheme of applying symmetries to groups, however, there has been no discussion on how the generators between each group interact. To see this however, it is beneficial to investigate a group that encapsulates all of these symmetries at once: the Galilean group.

We must remark at this point that we have ignored the idea of projective Hilbert spaces[7], and representations on these spaces, and thus the ideas in this section will be discussed informally. Thus if we apply the above scheme to the Galilean group¹⁰, we get a set of unitary and self-adjoint operators, satisfying the commutation relations of the associated Lie algebra. In fact, these relations are exactly the angular momentum, linear momentum, position, and Hamiltonian operators with the appropriate commutation relations including the canonical bracket[8]:

$$[q_i, p_j] = i\hbar\delta_{i,j}I$$

This scheme is very similar to creating a unitary representation of the Poincaré group to arrive at relativistic quantum mechanics. In fact, by using this group, there exists a natural mass operator that acts as a Casimir element, which commutes with every other operator associated with elements of the Lie algebra[7].

Now we discuss the pathologies of the spectrum of operators acting on infinite dimensional Hilbert spaces. We know that from 10, we can think of the spectrum as generalized eigenvalues and the spectral subspaces as generalized eigenspaces. However, this theorem has subtlety that eigenkets¹¹ need not necessarily be in the Hilbert space. This can be seen when looking at the eigenfunctions of the angular momentum operators which are the spherical harmonics $Y_l^m(\theta, \phi)$. While these functions are normalizable on the unit sphere, when we introduce the full Hilbert space, $L^2(\mathbb{R}^3, \mathbb{C})$, the square integral has the form:

$$\int_{\mathbb{R}^3} (Y_l^m(\theta, \phi))^* Y_l^m(\theta, \phi) d^3x = \int_0^\infty r^2 dr$$

which evidently diverges. This means that the spherical harmonics are in fact not in the Hilbert space $L^2(\mathbb{R}^3, \mathbb{C})$. This is even more clear if we look at the eigenkets of the position operator, which have the form of plane waves:

$$\int_{\mathbb{R}^3} ||e^{ikx}||^2 d^3x = \int_{\mathbb{R}^3} 1 d^3x$$

which evidently are not square integrable, and therefore do not live in the Hilbert space.

This in fact is not a violation of the theorems that we had built up through the previous sections, since in the statement of the unbounded spectral theorem (10), if we attempt to find a true discrete eigenvalue, we find that it forces the vector $\psi(x)$ to be zero. This issue only emerges if we try to

¹⁰ More precisely, the central extension of the Galilean group: the Bargmann group

¹¹ These are the vectors that span the spectral subspaces

force the spectral subspaces to be generated by eigenvectors. We do this to simplify our notation and to use Dirac's bra-ket notation in a consistent manner. Thus this contradiction is not intrinsic to the theory, but rather an artifact of our notation. This notational simplification can be made rigorously through rigged Hilbert spaces, a "larger" Hilbert space, where these eigenvectors live in, and generate the spectral subspaces of the operators [9]. We discuss this subtlety as it is a common pain point when discussing solutions to Schrödinger's equation for a free particle, and discussing why the eigenfunction is actually not a valid wave function.

VI. CONCLUSION

Through this construction, we developed a more intuitive understanding of where operators associated with physical observables come from. We found through the synthesis of the exponential map, Stone's theorem, and unitary representations, a method to easily associate continuous symmetries with self-adjoint operators.

Although we already omitted all proofs and many of the related theorems and propositions not core to our method, the mathematical background is still rather burdensome to go through. Perhaps an even larger simplification could be done to make this construction approachable to a student learning quantum mechanics for the first time. However, for a student of mathematics, this may be a simple way of developing the operators in quantum mechanics without too much unjustified assumption.

On a parallel note, we have begun to develop some ideas in the paper that may also be worthwhile to develop further in a pedagogical sense. In general, the current approach to teaching undergraduate quantum mechanics is very arbitrary and often made more confusing than it has to be. The difficulties lie in balancing the mathematical background required and the intuition of the approach. While the structure of this paper is rather unsuitable for a first introduction to quantum mechanics course, with some appropriate simplifications for some formal mathematical constructions, this approach can be framed in a manner that introduces the structure of quantum mechanics in an intuitive style.

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